

Sustaining Scalable Sustainability: Human-Centered Green Technology for Community-wide Carbon Reduction

Vikram Mohanty

vikrammohanty@acm.org
Bosch Research North America
Sunnyvale, California, USA

Hamed S. Alavi

h.alavi@uva.nl
University of Amsterdam
Amsterdam, Netherlands

Elizabeth Churchill

churchill@acm.org
Google
Mountain View, USA

Jingchao Fang

jcfang@ucdavis.edu
University of California, Davis
Davis, California, USA

Joaquin Salas

salas@ieee.org
Instituto Politécnico Nacional
Querétaro, Mexico

Charlene C. Wu

charlene.wu@tri.global
Toyota Research Institute
Los Altos, California, USA

Song Mi Lee-Kan

songmil@umich.edu
University of Michigan
Ann Arbor, Michigan, USA

Geneviève Patterson

patterson.genevieve@gmail.com
Ocean Vision AI
USA

David A. Shamma

aymans@acm.org
Toyota Research Institute
Los Altos, California, USA

ABSTRACT

Escalating global CO₂ emissions highlights the immediate need for scalable sustainable practices. Corporate and policy roles aside, there's a need for carbon-neutrality-based systems and practices to bridge the disconnect between actions and the perceived impact on the environment. This one-day workshop focuses on individuals and communities, advocating for human-centered tools to bridge this awareness-action gap. While Human-AI Interaction (HAI), cognitive science theories, and social computing tools have shown promise in various domains, their potential remains largely unexplored in the context of sustainability. This workshop aims to delve into these avenues for crafting tractable systems for effective, contextually relevant interventions and driving sustainable behaviors. By engaging multidisciplinary researchers, we aim to intertwine local insights with behavioral theory and technology, fostering intrinsic carbon literacy and a sustainability ethos, ensuring lasting and scalable impacts.

KEYWORDS

Carbon Neutrality, Carbon Literacy, CO₂, Eco-Feedback, Sustainability, Design Interventions, Human-Centered Design, Human-Centered AI, Persuasive Technology, Communities

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1 MOTIVATION

Global CO₂ emissions continue to rise, spurred by human activities across sectors like transportation, agriculture, and energy production [8, 29], underscoring a pressing need for sustainable practices. While corporations and policymakers have a pivotal role to play in reducing Greenhouse Gas (GHG) emissions [6, 28], this **workshop focuses on individuals and communities**, acknowledging them as critical agents of change [12, 41]. Private households account for and have the potential to influence approximately two-thirds of global emissions [35]. However, research indicates that individuals often do not associate their actions with climate change mitigation, instead attributing the primary responsibility to larger impersonal entities like industries [42]. Additionally, individuals are often unaware of the relative impact of carbon-emitting and mitigating activities [17], which can greatly differ by community and locality. For instance, while rehabilitating mangroves may be more locally appropriate and involve and benefit local people to a greater extent, individual efforts might be misdirected to lower-impact actions, such as recycling [14, 40]. Such misconceptions can not only undermine the effectiveness of policies but also highlight the untapped potential of local capacities and resources to substantially reduce CO₂ emissions. To bridge this gap, fostering *carbon literacy*—a nuanced understanding of carbon impacts and actionable steps individuals and communities can take to mitigate emissions—is paramount to catalyzing transformative, sustainable change [5, 19, 22, 38].

Previous experiments with pro-environmental technological interventions demonstrate a promising outlook for nurturing a sustainability culture from the grassroots. For instance, utilizing real-time eco-feedback information (i.e., informing users about their resource consumption) in areas such as food consumption, energy usage, transit, charging, and waste disposal can encourage green consumer choices [11, 16, 31, 32, 37, 43, 44]. Leveraging machine

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learning (ML) and data visualization techniques has enabled intricate analyses of ecological data and resource consumption information sourced from IoT sensors, application logs, etc [21, 25, 26, 36]. The data-driven approach can reveal hidden patterns, discern trends, and help in identifying key focal areas with significant environmental or resource impact, thereby facilitating more informed decision-making towards a sustainable future. However, scaling these technological interventions for broader sustainable behavior change is impeded by key gaps in human-centered design—primarily concerning a lack of granularity and adaptability to fit the lived contexts.

Many existing interventions have yet to achieve long-term, large-scale voluntary adoption, indicating a disconnect between designed solutions and the real-world practices of individuals and communities [3]. The prevalent *one-size-fits-all* design approach often overlooks diverse needs and values of individuals and distinct contexts of local communities, limiting the adoption and effectiveness of these interventions [2, 15, 34]. For instance, fact-based information provision (e.g., delivering statistics, general guidelines, or one-time carbon literacy education) has had the smallest effect on sustainable behavior change compared to other types of interventions [3]. This may likely be due to an inability to account for variances in individuals' beliefs, cognitive styles, psychological tendencies (e.g., reactance to cognitive dissonance and guilt-inducing realities), and limited behavioral control over environmental constraints [7, 30]. Moreover, community capacity can also vary widely in terms of inclusivity and cohesion for undertaking collective initiatives, its relationships with other governance actors, and the control over enabling resources [40].

Despite the significant challenges in fostering general public carbon literacy, **human-centered design in the fields of Human-Computer Interaction (HCI) and Artificial Intelligence (AI)** is uniquely positioned to navigate and adapt to individual and local complexities. HCI, bridging various disciplines including information science, psychology, cognitive science, and behavioral economics, has spearheaded advancements in personalized human augmentation. It provides a theoretically-grounded roadmap for developing interventions that resonate with local contexts, cultures, and beliefs, avoiding potential backfires from the 'one-size-fits-all' pitfall [15]. Prior HCI endeavors have also excelled in community engagement within domains like urban planning, policymaking, and citizen science [18, 27], and notably, including marginalized voices [4, 9]. Building on this, there emerges a compelling opportunity to leverage similar methods to **understand the sustainability needs and practices of diverse local communities**, while concurrently designing technology that empowers them with actionable, meaningful insights, systems, and tools.

Human-AI Interaction (HAI) is seeing persuasive outcomes in domains like health and financial decision-making [20], signaling its potential to influence sustainable behaviors. Using Large Language Models (LLMs) as tools present opportunities to craft pro-environmental messages and behavioral support, tailored to demographic and psycho-social variances [1], thereby bridging the gap between awareness and actionable insights at both individual and community levels. Noteworthy is its recent extension to simulate social interactions with generative agents [33], which can assist in exploring diverse intervention scenarios based on historical data at a reduced cost. Similarly, social computing tools also possess

untapped potential, be it for enabling communities to collectively deliberate on sustainable decisions [23] or for fostering behaviors like eco-driving or energy savings [13, 31, 39].

The pressing challenges of fostering carbon literacy and active community participation in *actionable sustainability initiatives* underscore the need for **innovative methods, tangible prototypes, and actionable tools** that can be implemented and deployed in the present. While policy and industry adaptations are essential, they often move at a methodical pace, highlighting the urgency of grassroots, bottom-up solutions to catalyze immediate change. While not exhaustive, our perspective highlights a vast space of human-centered green technologies, with a pivotal benefit to local communities. In this workshop, we invite researchers across various disciplines to collaboratively explore, expand, and harness this potential.

2 WORKSHOP GOALS AND GUIDING QUESTIONS

Our primary goal is to establish a forward-thinking, ready-to-deploy research agenda for designing sustainability technologies that emphasize community-level adoption and adaptation. In particular, we focus on roadmaps and systems ripe for immediate testing and/or deployment. This bottom-up approach uniquely positions us to integrate humans directly into the design loop, leveraging the expansive expertise of HCI researchers in human-centered design. The research themes we will explore include:

- **Individuals and Communities as Agents of Change:** How can we empower individuals and communities to become the driving force behind sustainable practices?
- **Technologies for Sustainability:** How can we develop tractable systems and tools that enable pragmatic decision-making and are scalable to promote broader sustainability efforts?
- **Diverse and Large-scale Adoption:** How can we ensure our interventions are inclusive, culturally sensitive, and adopted on a large scale?

Within these overarching themes, we will spark discussions and debates on various aspects vital to the design of human-centered green technology. Some pivotal questions include:

- Design strategies for long-term attitude shift towards sustainable practices.
- Effective value-driven communication of carbon information.
- Identify technology and interventions that can be deployed now.
- Context-aware designs that move beyond a *one-size-fits-all* model.
- Leveraging Human-AI Interaction for effective eco-awareness and persuasion.
- Behavioral science-driven methods for effective adoption of technology interventions.
- Amplifying individual efforts through social computing and group dynamics within communities.

This workshop aims to engage researchers from diverse backgrounds, including AI, HCI, behavioral sciences, social sciences, and

beyond, to collectively delve into the development of sustainable technologies that address the crucial gaps. Through this workshop, we aim to emphasize research systems and directions that look at **a harmonious relationship between communities, technology, and sustainable practices**. By engaging communities from the outset, we can not only aim to understand their unique needs and experiences, but also design solutions to facilitate a two-way exchange of indigenous sustainable practices and technological innovations. This communal insight, coupled with **behavioral theory-driven design**, can shape the development of **culturally resonant interventions**. This integrative approach, combining local wisdom with advanced technology and theory, sets a fertile ground for **fostering carbon literacy through both practical engagement and education**. Our ultimate objective is to compile a concrete roadmap toward advancing human-centered green technology in a bottom-up approach, and cultivate a deeper understanding and commitment towards sustainability, making it an intrinsic part of individual and community ethos.

3 ORGANIZERS

Vikram Mohanty is a Postdoctoral Researcher at Bosch Research and Technology Center, USA. His research focuses on human-AI interaction, online communities, and crowdsourcing. He received his Ph.D. from the Department of Computer Science at Virginia Tech. As part of his research, he built Civil War Photo Sleuth, an online platform that combines crowdsourced human expertise and AI to identify unknown portraits from the American Civil War era. He has also worked on exploring effective eco-feedback interventions for encouraging users to make sustainable choices in vehicular contexts, such as ridesharing, car rentals, and charging. Vikram has worked as a research intern at VSCO, Dataminr, and Toyota Research Institute (TRI). Prior to his PhD, he completed his undergraduate from the Indian Institute of Technology (IIT), Kharagpur.

Jingchao Fang is a Ph.D. candidate in the Department of Computer Science at the University of California, Davis. Her research interest lies at the intersection of HCI, CSCW, and Educational Technology. She focuses on studying and designing interactive systems that facilitate knowledge-based communication and collaboration. During her internship at Toyota Research Institute, she explored the potential of leveraging generative AI to design green interventions that promote sustainable lifestyles. She obtained her bachelor's degrees in Computer Science and Mathematics from the University of Minnesota, Twin Cities.

Song Mi Lee-Kan is a PhD Candidate at the University of Michigan's School of Information. Her research bridges information literacy and human-computer interaction, with an emphasis on the psychosocial factors that lead to the misuse and abuse of information. Song's research currently centers on interventions aimed at bolstering interest in carbon emissions information, a vital precursor to achieving carbon literacy and the adoption of carbon-neutral technologies. Song holds master's degrees in Library and Information Science from both Indiana University Bloomington and Yonsei University in South Korea.

Hamed S. Alavi is an assistant professor and a founding member of the Digital Interactions Lab at the University of Amsterdam. His research interests lie at the crossroads of Human-Computer Interaction and the scientific domains of Architecture and Urban Design. Particularly, he seeks to investigate how the incorporation of new forms of intelligence and interactivity in buildings and cities can reshape the relationship between humans and their environments. Such relations are highly relevant to topics surrounding sustainability concerns since over one-fifth of global energy use is attributed to maintaining indoor environmental parameters within the standard comfort ranges. Hamed has previously organized several workshops at CHI. He was also the guest editor-in-chief for a TOCHI special issue in 2019.

Joaquin Salas is a professor in the field of Computer Vision at Instituto Politécnico Nacional. Member of the Mexican National Research System, his research interests include monitoring natural systems using visual perception and remote sensing platforms. Salas received a Ph.D. in computer science from ITESM, México. He has been a visiting scholar at MIT, Stanford, Duke, Oregon State, Xerox PARC, the Computer Vision Center, and the ÉNIST de Bretagne. Salas was a Fulbright scholar for the US State Department. He has been invited as editor for Elsevier Pattern Recognition and Pattern Recognition Letters. For his services at the Instituto Politécnico Nacional, he received the Lázaro Cárdenas medal from the President of Mexico.

Geneviève Patterson is a computer vision and HCI researcher pushing the boundaries of AI-science collaboration with the Ocean Vision AI initiative at the Monterey Bay Aquarium Research Institute. Previously, she was Head of Applied Research at the popular photography app VSCO, which acquired her computational cinematography startup in 2020. Recently, she was part of the team awarded the 2023 PAMI Mark Everingham Prize for her role in the Common Objects in Context Challenge, which helped to drive the recent revolution in image detection and segmentation. Her interests include video understanding, visual attribute discovery, human-in-the-loop systems, fine-grained object recognition, AI for climate science, and active learning. Geneviève has previously served as the Director of Projects and Partnerships at Climate Change AI and edited a Special Issue on Artificial Intelligence for Sustainability in the IEEE Latin America Transactions. She received her PhD from Brown University in 2016.

Elizabeth Churchill is a Senior Director of UX at Google and Co-Chair of Google's UX Leadership Council. She has built research teams at Google, eBay, Yahoo, PARC, and Fuji Xerox. Her current focus is on the design of effective designer and developer tooling for a new operating system, Fuchsia.

Charlene C. Wu is a behavioral science leader bridging academic and technology sectors to advance research in human-centered AI. She is currently the Director of the Human-Centered AI (HCAI) division at the Toyota Research Institute (TRI). The HCAI division aims to develop AI systems to augment human decision-making by leveraging research at

the intersection between behavioral science, machine learning, and human-computer interaction. Before TRI, Charlene founded the applied behavioral data science disciplines at Uber and Airbnb, where her team applied behavioral science research and methodologies to generate novel hypotheses and prototype solutions to guide global business strategy. Charlene is passionate about discovering and pursuing novel opportunities where behavioral science can have an outsized impact. Charlene has been a speaker at the World Economic Forum, Behavioral Science Summit, and Women in Analytics conferences and serves as a mentor for Data Science for All and a strategic advisor for both early-stage startups and publicly traded companies. She holds Bachelor's degrees in economics and psychology from UCLA and Master's and Ph.D. degrees in psychology from Stanford University.

David A. Shamma is a distinguished industry scientist researching people, artificial intelligence, and media experiences. His current research focuses on Carbon Neutrality at Toyota Research Institute's Human-Centered AI division. He has worked on Edge AI and future of work at FX Palo Alto Laboratory, AI sensors for wearable fashion at Centrum Wiskunde & Informatica (CWI), HCI + AI as a director of research at Yahoo Labs/Flickr, and UXR for remote knowledge sharing at NASA's Center for Mars Exploration. He is a Distinguished Member of the ACM, a Senior Member of IEEE, and holds over 30 patents. Ayman holds a Ph.D. in Computer Science from Northwestern University and completed his M.S./B.S. at the IHMC at UWF.

4 PLANS TO PUBLISH WORKSHOP PROCEEDINGS

We plan to host all the accepted submissions on the workshop website¹ and potentially post a workshop summary on arXiv.

5 WORKSHOP FORMAT

While we are hopeful that many will be able to travel to CHI 2024 in Hawaii, we acknowledge that it may not be feasible for everyone. In a bid to be inclusive, we have structured our one-day workshop to operate in a hybrid format. We will establish a chat infrastructure and provide a Zoom room for synchronous participation, enabling attendees to dial in for discussions, critiques, or remote demo presentations. The organizers bring substantial experience in remote collaboration tools to the table². By adopting a hybrid model, we aim to address both sustainability and inclusion concerns, reducing financial and environmental barriers associated with travel and ensuring a broader spectrum of participation. We also plan on including some support for asynchronous engagement (as described in Section 6).

6 ACCESSIBILITY AND ASYNCHRONOUS ENGAGEMENT

For our accepted authors, we intend to establish a Discord or Slack server to foster early conversations among attendees. All accepted

submissions will be shared with participants in advance. During the workshop, communication will be facilitated through recorded Zoom calls and the Discord/Slack servers, ensuring accessibility for later reference. Additionally, panel/keynote sessions and presentations will be recorded and made available for participants to review at their convenience. We will also designate someone (either from the organizer team or a volunteer) to take notes, photos, and recordings of the workshop activities, that will be made available for everyone.

Accessibility is of critical importance and we will ensure our workshop is accessible to all those who attend. The organizing committee has members who are accessibility researchers or have a proven track record of building accessible systems. Some things to note: we will request that all presentation materials be made available before the workshop, which will include correctly tagged PDFs or accessible slide decks or captioned videos. We will request CART services as needed for in-situ or hybrid. We will ensure that SIGACCESS-approved technologies are used and SIGCHI's WCAG2.1 AA or AAA standard is followed. This is just to name a few points. If we need to secure additional accessibility features at cost, we are aware of the SIGCHI requesting procedures if CHI cannot cover the cost (as per SIGCHI's bylaws).

7 PRE-WORKSHOP PLANS

Upon acceptance, we will first set up a dedicated website (<https://sustainingscalablesustainability.wordpress.com/>) to promote the workshop and its call for proposals. This platform will not only serve as an information hub for prospective attendees but will also feature workshop contributions, such as accepted papers, fostering ongoing engagement. Beyond the website, we will also directly reach out to our colleagues in industry and academia, promote the workshop on social media, and advertise via email distribution to related conferences and academic institutions, tapping into networks like AutoUI, Climate Change AI, CogSci, IUI, CHI, NeurIPS, HCOMP, Collective Intelligence, and CSCW, among others. Given the interdisciplinary nature of our organizing committee's research, we will proactively engage a wide array of communities, including HCI, AI/ML, Behavioral Science, Architecture, Automobile, Environmental Science, Sustainable HCI, etc. We also plan to secure some small funding for the workshop to cover some expenses that might arise (i.e., dinners, honorariums, etc.) or to help cover some costs for attending students. For our accepted authors, we intend to set up a Discord or Slack server to kickstart discussions among attendees. In preparation for certain workshop activities, we will provide them with essential materials and notes to ensure they are well-prepared and can actively engage in the planned activities.

8 WORKSHOP ACTIVITIES

We anticipate hosting 20–25 participants. For the workshop, we are planning a morning session that starts with a kickoff talk, followed by a lightning round of introductions, and a roundtable session. In the afternoon, we plan on conducting a 1.5–2 hrs long “speculative hackathon” session.

¹<https://sustainingscalablesustainability.wordpress.com/>

²<https://medium.com/sigchi/making-the-first-hybrid-chi-in-2022-11636b43be0a>

This “speculative hackathon” session will combine speculative design fiction’s critical *design thinking* that envisages future scenarios with the *tangible results* of classic hackathons, allowing participants to envision sustainable futures while addressing today’s pragmatic challenges. The organizers will curate realistic scenarios highlighting challenges that hinder the translation from theoretical knowledge in carbon literacy to everyday practices (e.g., long charging time and infrastructural demands of electric vehicles, cost trade-offs of sustainable choices [10, 24, 31]). Participants will be provided with a suite of tools—from prototypes, legos, AI-infused interfaces, and visualizations to LLMs and design cards—and encouraged to generate actionable insights, conceptual ideas, and prototypes that address these sustainability challenges in a pragmatic manner.

We envision the one-day workshop to take the following structure:

- Welcome Greetings: The organizers will introduce themselves and the workshop’s objectives and the schedule for the day.
- Opening Talk / Panel: We are currently in discussions with local groups in Hawaii and the Monterey Bay Aquarium Research Institute (MBARI) regarding a potential panel discussion or a keynote.
- Coffee Break
- Lightning Talks: (participants will introduce themselves, their work, their lab/organization, their submissions, interesting offline discussions moderated by the organizers)
- Lunch
- Roundtable Discussion
- Break
- Speculative Hackathon
- Hackathon Presentation: Teams present their derived outcomes, takeaways, or speculative solutions, followed by discussion and feedback from other participants and organizers.
- Wrap-up Summary
- Dinner

9 POST-WORKSHOP PLANS

After the workshop, we plan to leave the aforementioned chat system in place for our attendees to continue the conversation both during and beyond the conference. We will aim to submit an article to ACM Interactions from our workshop’s proceedings. We will also plan on organizing a special issue in the ACM Journal on Computing and Sustainable Societies. Our stretch goal will be to use this inaugural CHI workshop as a springboard to move the workshop in the following years to other conferences and disciplines: NeurIPS, CogSci, and RecSys to name a few. Through these efforts, we will aim to expand the interdisciplinary nature of this work and establish the CHI conference as the cornerstone for this research in Human-Centered Green Tech.

10 CALL FOR PARTICIPATION

In alignment with the “Sustaining Scalable Sustainability” theme, this workshop seeks collaborative endeavors to build sustainable technology that deeply resonates with individuals and communities, guiding them toward sustainable practices and carbon literacy.

We invite interdisciplinary researchers and practitioners from various disciplines such as AI, HCI, psychology, and social sciences to present innovative, community-engaged, and theory-driven approaches, advancing us from mere awareness to actionable sustainability steps. The workshop will be a one-day hybrid event.

We invite submissions of all forms, but not limited to:

- Position statements and papers (2–4 pages) in the new ACM master article template format.
- Slide decks
- Video demos
- Hack ideas
- Systems, prototypes (including Figma), or conceptual ideas

Possible submission topics, aligned with our theme of actionable and scalable sustainability, may include (but are not limited to):

- Data visualizations
- Human-AI interactive tools, which may include generative AI
- Understanding and empowering local communities
- Speculative future scenarios
- Cognitive Science theory-driven interventions.
- Large scale analysis of ecological/consumption data

Participants will send their submissions to humancenteredgreentech@gmail.com. The submissions will be evaluated on the basis of relevance to the workshop’s theme. Accepted contributors will prepare a 5-minute workshop presentation and join our chat server for discussing one another’s submissions. We will publish the accepted submissions on the workshop website (<https://sustainingscalablesustainability.wordpress.com/>), along with other relevant information about the workshop.

At least one author of each accepted paper must register for the workshop and at least one day of the conference and attend the workshop.

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